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Inventor(s)

JOUKAN; Seimizu

### DISPLAY CONTROL DEVICE, DISPLAY CONTROL SYSTEM, DISPLAY CONTROL METHOD, AND RECORDING MEDIUM

#### Abstract

A display control device used in a vehicle provided with a display that includes a plurality of display regions. The display control device includes a display region manager that manages the plurality of display regions. When the display region manager obtains, from an external terminal, a display request for displaying display data on a display region included in the plurality of display regions, the display region manager creates a display region table collectively indicating display region information on at least one display region among the plurality of display regions, and outputs the display region table to the external terminal.

**Inventors:** JOUKAN; Seimizu (Osaka, JP)

**Applicant:** Panasonic Automotive Systems Co., Ltd. (Kanagawa, JP)

**Family ID:** 85782407

**Assignee:** Panasonic Automotive Systems Co., Ltd. (Kanagawa, JP)

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation of U.S. patent application Ser. No. 18/615,716, filed Mar. 25, 2024, which is a continuation of PCT International Patent Application No. PCT/JP2022/034048, filed Sep. 12, 2022, which claims priority to Japanese Patent Application No. 2021-160543, filed Sep. 30, 2021. The entire contents of each of the above-mentioned documents is incorporated herein by reference.

### FIELD

[0002] The present disclosure relates to a display control device, a display control system, a display control method, and a non-transitory computer-readable recording medium having a computer program recorded thereon.

### BACKGROUND

[0003] Patent Literature 1 discloses a conventional on-vehicle system that causes an on-vehicle device and a mobile terminal device to work together, and allocates and displays application software information items in the on-vehicle device to a plurality of display regions.

[0004] Patent Literature 2 discloses a conventional apparatus that receives a display signal from an inputter and causes one or more of a plurality of displays to display a selected region in accordance with an input received through a user input device.

### CITATION LIST

Patent Literature

[0005] PTL 1: Japanese Patent No. 5920474 [0006] PTL 2: Japanese Patent No. 5725259

### SUMMARY

[0007] However, the on-vehicle system according to PTL1 and the apparatus according to PTL2 can be improved upon.

[0008] In view of this, the present disclosure provides a display control device, a display control system, a display control method, and a non-transitory computer-readable recording medium on which a program is stored, each capable of improving upon the above related arts.

### Solution to Problem

[0009] In order to achieve the above object, in accordance with an aspect of the present disclosure, a display control device for use in a vehicle provided with a plurality of displays each including one or more display regions includes: a display region manager that manages a plurality of display regions that are the one or more display regions included in each of the plurality of displays, wherein when the display region manager obtains, from an external terminal, a display request for displaying display data on a display region included in the plurality of display regions, the display region manager: creates a display region table collectively indicating display region information on at least one display region among the plurality of display.

[0010] General or specific aspects of the present disclosure may be implemented to a system, a device, a method, an integrated circuit, a computer program, or any given combination thereof.

## Description

### BRIEF DESCRIPTION OF DRAWINGS

[0012] These and other advantages and features of the present disclosure will become apparent from the following description thereof taken in conjunction with the accompanying drawings that illustrate a specific embodiment of the present disclosure.

[0013] FIG. 1 is a diagram illustrating an example of a display control system and a vehicle according to an embodiment.

[0014] FIG. 2 is a block diagram illustrating an example of the display control system according to the embodiment.

[0015] FIG. 3 is a diagram illustrating an example of a display region table.

[0016] FIG. 4 is a sequence diagram illustrating an example of a processing operation between the display control system and an external terminal according to the embodiment.

[0017] FIG. 5 is a sequence diagram illustrating an example of a processing operation between the display control system executing an exclusive display mode and the external terminal.

[0018] FIG. 6 is a sequence diagram illustrating an example of a processing operation between the display control system executing a message display mode and the external terminal.

[0019] FIG. 7 is a sequence diagram illustrating an example of a processing operation between the display control system executing an overlap display mode and the external terminal.

[0020] FIG. 8 is a sequence diagram illustrating an example of a processing operation between the display control system executing an event driven mode and the external terminal.

### DESCRIPTION OF EMBODIMENT

[0021] Hereinafter, certain exemplary embodiments will be described in detail with reference to the accompanying Drawings.

[0022] The following embodiments are general or specific examples of the present disclosure. The numerical values, shapes, materials, elements, arrangement and connection configuration of the elements, steps, the order of the steps, etc., described in the following embodiments are merely examples, and are not intended to limit the present disclosure. Among elements in the following embodiments, those not described in any one of the independent claims indicating the broadest concept of the present disclosure are described as optional elements.

[0023] Note that the respective figures are schematic diagrams and are not necessarily precise illustrations. Therefore, the scales or the like applied in the figures are not necessarily unified. Additionally, components that are essentially the same share like reference signs in the figures. Accordingly, overlapping explanations thereof are omitted or simplified.

#### Embodiment

##### <Configuration: Display Control System 2>

[0024] FIG. 1 is a diagram illustrating an example of display control system 2 and vehicle 1 according to an embodiment. FIG. 2 is a block diagram illustrating an example of display control system 2 according to the embodiment.

[0025] As illustrated in FIG. 1 and FIG. 2, display control system 2 is provided in vehicle 1. Vehicle 1 is provided with a plurality of displays 10. When display control system 2 obtains a terminal-application image from external terminal 100, display control system 2 can display the terminal-application image obtained on display 10 selected to be used by external terminal 100 from among the plurality of displays 10.

[0026] External terminal 100 is a terminal capable of communicating with display control system 2 in a wired or wireless manner. External terminal 100 is, for example, a personal computer, or a

portable terminal such as a smartphone or a tablet terminal. When external terminal **100** is to use one or more display regions **10a** included in a plurality of display regions **10a**, external terminal **100** can issue, to display control system **2**, a display request for displaying display data on one or more display regions **10a** included in the plurality of display regions **10a**. When issuing the display request, external terminal **100** can also notify display control system **2** of one or more identifiers of the one or more display regions **10a** selected to be used.

[0027] After issuing the display request, external terminal **100** can issue, to display control system **2**, a use request for permission to use the one or more display regions **10a** that are selected to use. When display control system **2** permits the use request, external terminal **100** can cause one or more displays **10** of display control system **2** to display the display data such as the terminal-application image.

[0028] Specifically, when the use request is permitted, external terminal **100** transmits display information including information to be displayed on the one or more display regions **10a** to display control system **2**. The display information includes the terminal-application image. The display information will be described later in detail. The terminal-application image is an image of an operating system, an image of a navigation application, an image of an audio application, or the like in external terminal **100**.

[0029] External terminal **100** may transmit a plurality of display information items to display control system **2**, and display control system **2** may cause respective display regions **10a** to display images rendered.

[0030] Display control system **2** includes the plurality of displays **10**, first storage **21**, communicator **22**, and display control device **30**. Display control system **2** may further include external terminal **100**.

[Display **10**]

[0031] The plurality of displays **10** are displays **10** for a vehicle such as a center display, a head-up display (HUD), an instrument cluster (IC) display, and a passenger display provided in vehicle **1**. When vehicle **1** travels backward, each of the plurality of displays **10** can display an image captured by an image capturing unit, thus functioning as a rear-view monitor.

[0032] Displays **10** each include one or more display regions **10a**. That is, a screen of each of the plurality of displays **10** is provided with one or more display regions **10a** each of which displays an image. Therefore, a screen of one display **10** may be set with one display region **10a** or may be set with a plurality of display regions **10a**. In the present embodiment, two displays **10** are exemplified. One of the two displays **10** is an IC display, and the IC display device is set with display regions A and B. The IC display is a display disposed in an instrument panel. The IC display is suitable for displaying measuring instrument information, turn-by-turn information, traffic information, disaster information, driving information, or the like. The other of the two displays **10** is a center display, and the center display is set with display regions C, D, and E. The center display is, for example, a display disposed between a driver's seat and a passenger seat. The center display is suitable for displaying navigation information, media message information, or the like. In the present embodiment, the display regions A, B, C, D, and E may be collectively and simply referred to as display regions **10a**. Sizes, positions, and the like of display regions **10a** may be configured to be changed freely.

[First Storage **21**]

[0033] First storage **21** stores vehicle applications, a vehicle rendering command, and the like. Here, the vehicle applications include an operating system for a computer, a navigation application, an audio application, and the like for displaying images on the plurality of displays **10**. The vehicle rendering command is a command for requesting display control device **30** to render a vehicle-application image from the vehicle application. The vehicle-application image is an image of an operating system, an image of a navigation application, an image of an audio application, or the like in external terminal **100**.

[0034] First storage **21** includes a primary storage device such as a random access memory (RAM) and a read only memory (ROM). First storage **21** may include a secondary storage device such as a hard disk drive (HDD) or a solid state drive (SSD) and may include a tertiary storage device such as an optical disk or an SD card.

[Communicator **22**]

[0035] Communicator **22** is, for example, a communication interface provided in vehicle **1** and enables communication between vehicle **1** and external terminal **100**. Communicator **22** receives, from external terminal **100**, a display request, a use request, a terminal-application image, and the like. Communicator **22** transmits, to external terminal **100**, a permission for using of display region **10a**, an output request for outputting of display information, or the like. Communicator **22** may be included in constituent components of display control device **30**.

[Display Control Device **30**]

[0036] Display control device **30** is used in vehicle **1** provided with a plurality of displays **10** each including one or more display regions **10a**. When display control device **30** obtains a terminal-application image from external terminal **100** and obtains a vehicle-application image from first storage **21**, display control device **30** generates images to be displayed on display regions **10a** of displays **10** based on the terminal-application image and the vehicle-application image obtained. Thus, display control device **30** can cause displays **10** to display the images generated.

[0037] Specifically, display control device **30** includes second storage **31**, display region manager **32**, display controller **33**, control event generator **35**, and display screen generator **34**.

[Second Storage **31** of Display Control Device **30**]

[0038] Second storage **31** stores a management table. The management table is a table created beforehand. The management table is, for example, a table including display region information on all display regions **10a** set in vehicle **1**.

[0039] Second storage **31** includes a primary storage device such as a RAM and a ROM. Second storage **31** may include a secondary storage device such as an HDD or an SSD and may include a tertiary storage device such as an optical disk or an SD card.

[0040] Display control device **30** need not include second storage **31**. In this case, display control device **30** may obtain the management table from an external device.

[Display Region Manager **32** of Display Control Device **30**]

[0041] Display region manager **32** manages the plurality of displays **10**, that is, the plurality of display regions **10a**. Specifically, when display region manager **32** obtains a display request from external terminal **100**, display region manager **32** reads the management table from second storage **31** and creates a display region table as illustrated in FIG. **3** using the management table read. FIG. **3** is a diagram illustrating an example of a display region table.

[0042] Here, the display region table collectively indicates display region information that is information on at least one display region **10a**. That is, the display region table includes at least one display region information item. The display region information includes, for each of the at least one display region **10a**, an identifier, a position, a size, a display mode, presence or absence of an input and an output for information, and a use condition. Examples of the information include a vehicle-application image, a terminal-application image, a terminal rendering command, a vehicle rendering command, display information, and an event.

[0043] The identifier of display region **10a** is a unique identifier with which display region **10a** can be identified. The position of display region **10a** is a position of display **10** disposed in vehicle **1** and a position at which display region **10a** is displayed on a screen of display **10** disposed in vehicle **1**. The size of display region **10a** is a size of display region **10a** within which display region **10a** can be displayed on the screen of display **10**. The display mode of display region **10a** includes an exclusive display mode, a message display mode, an overlap display mode, and an event driven mode. The modes will be described later in detail. The presence or absence of an input and an output for information is, for example, presence or absence of a terminal for an input and an output

for information. The use condition of display region **10a** has a setting for an arbitrary application and a setting for an authenticated application. For the arbitrary application, such a setting that any external terminal **100** can use display region **10a** is provided. For the authenticated application, such a setting that only external terminal **100** having a usage right can use display region **10a** is provided. For example, by registering external terminal **100** having the usage right beforehand, external terminal **100** registered can use display region **10a** to which the authenticated application is set.

[0044] For example, identifier A of display region **10a** is associated with a position of display region **10a** (Cluster center), a size of display region **10a** (600×400), a display mode of display region **10a** (Exclusive display mode), presence or absence of an input and an output for information (Absent), and a use condition of display region **10a** (Arbitrary application).

[0045] For example, identifier B of display region **10a** is associated with a position of display region **10a** (Cluster bottom), a size of display region **10a** (800×500), a display mode of display region **10a** (Event driven mode), presence or absence of an input and an output for information (Absent), and a use condition of display region **10a** (Authenticated application).

[0046] For example, identifier C of display region **10a** is associated with a position of display region **10a** (IVI top), a size of display region **10a** (1200×200), a display mode of display region **10a** (Message display mode), presence or absence of an input and an output for information (Absent), and a use condition of display region **10a** (Arbitrary application).

[0047] For example, identifier D of display region **10a** is associated with a position of display region **10a** (IVI Center), a size of display region **10a** (1200×1000), a display mode of display region **10a** (Event driven mode), presence or absence of an input and an output for information (Present), and a use condition of display region **10a** (Arbitrary application).

[0048] For example, identifier E of display region **10a** is associated with a position of display region **10a** (IVI Center), a size of display region **10a** (300×300), a display mode of display region **10a** (Overlap display mode), presence or absence of an input and an output for information (Absent), and a use condition of display region **10a** (Authenticated application).

[0049] It should be noted that the display region table illustrated in FIG. 3 is merely an example. The display region table is not limited to the data described above.

[0050] Display region manager **32** may extract only one or more display regions **10a** allowed to be used to create the display region table or may extract display regions **10a** selected to be used by external terminal **100** to create the display region table.

[0051] For example, display region manager **32** may extract only one or more display regions **10a** allowed to be used from the plurality of display regions **10a** in the management table in second storage **31** and may create the display region table collectively indicating only the display region information on the one or more display regions **10a** extracted. That is, display region manager **32** may create a display region table from which one or more display regions **10a** that are not allowed to be used when a display request is issued, are excluded from the plurality of display regions **10a**.

[0052] When display region manager **32** obtains, from external terminal **100**, one or more identifiers of one or more display regions **10a** selected to be used by external terminal **100** together with a display request, display region manager **32** may extract one or more display regions **10a** selected to be used by external terminal **100** from the plurality of display regions **10a** in the management table in second storage **31** and may create a display region table collectively indicating the display region information on the one or more display regions **10a** extracted. That is, when display region manager **32** obtains one or more identifiers of one or more display regions **10a** selected to be used by external terminal **100** when a display request is issued, display region manager **32** may create a display region table collectively indicating display region information on all display regions **10a** that satisfy selection of using by external terminal **100**.

[0053] Next, after creating the display region table, display region manager **32** outputs the display region table created as a response to the display request from external terminal **100**, to external

terminal **100** via communicator **22**.

[0054] After outputting the display region table to external terminal **100**, display region manager **32** obtains, from external terminal **100**, a use request for permission to use one or more display regions **10a** selected to be used by external terminal **100** based on the display region table. That is, after outputting the display region table created to external terminal **100**, display region manager **32** obtains, as a response to the output, a use request for permission to use the one or more display regions **10a** selected by external terminal **100** based on the display region table, from external terminal **100**. In this case, display region manager **32** determines whether or not the one or more display regions **10a** requested by the use request are allowed to be used. Here, when determining that the one or more display regions **10a** are allowed to be used, display region manager **32** permits external terminal **100** for which the one or more display regions **10a** are determined to be allowed to be used, to use the one or more display regions **10a** selected by external terminal **100**, that is, permits an application of external terminal **100** to use the one or more display regions **10a**. When determining that the one or more display regions **10a** are not allowed to be used, display region manager **32** does not permit external terminal **100** for which the one or more display regions **10a** are determined not to be allowed to be used, to use the one or more display regions **10a** selected by external terminal **100**, that is, does not permit the application of external terminal **100** to use the one or more display regions **10a**.

[0055] Here, a case where display region manager **32** permits external terminal **100** to use one or more display regions **10a** will be simply exemplified.

[0056] For example, display region manager **32** may determine that one or more display regions **10a** are allowed to be used in response to a use request that is obtained first. Display region manager **32** then may permit external terminal **100** that has issued the use request for which the one or more display regions **10a** are determined to be allowed to be used, to use one or more display regions **10a** selected by external terminal **100**. The use request obtained first means, for example, a use request that is obtained first since a time point at which display region manager **32** first transmits the display region table.

[0057] Display region manager **32** may determine, in response to the use request, that one or more display regions **10a** are allowed to be used only by external terminal **100** to which a usage right has been granted beforehand. Display region manager **32** then may permit external terminal **100** that has issued the use request for which the one or more display regions **10a** are determined to be allowed to be used, to use one or more display regions **10a** selected by external terminal **100**. Here, external terminal **100** to which a usage right has been granted means that the usage right has been granted to a terminal application that is an application used by external terminal **100**.

[0058] Display region manager **32** may determine that one or more display regions **10a** are allowed to be used in response to a use request obtained from a device to which a preferential usage right has been granted beforehand. Display region manager **32** then may preferentially permit the device that has issued the use request for which the one or more display regions **10a** are determined to be allowed to be used, to use display regions **10a** selected by the device to which a preferential usage right has been granted. Here, the device includes a device provided in vehicle **1** other than external terminal **100**, display control device **30**, and display control device **30**. The device to which the preferential usage right has been granted means that an application used by the device has the preferential usage right.

[0059] In a case where display region manager **32** obtains a plurality of use requests, display region manager **32** may determine, in response to a use request for display region **10a** associated with a use that is most similar to a use of a display mode associated with display region **10a** among the plurality of use requests, that display region **10a** is allowed to be used. Display region manager **32** then may permit external terminal **100** that has issued the use request for which the one or more display regions **10a** are determined to be allowed to be used, to use one or more display regions **10a** selected by external terminal **100**.

[0060] Display region manager **32** may determine, in response to a use request obtained from external terminal **100** having a highest priority degree among a plurality of use requests obtained from a plurality of external terminals **100**, that one or more display regions **10a** are allowed to be used. Display region manager **32** then may permit external terminal **100** that has issued the use request for which the one or more display regions **10a** are determined to be allowed to be used, to use one or more display regions **10a** selected by external terminal **100**. Such a priority degree with which one or more display regions **10a** are preferentially allowed to be used is set beforehand to each of the plurality of external terminals **100** and each of applications of the plurality of external terminals **100**.

[0061] After display region manager **32** permits external terminal **100** to use the one or more display regions **10a** in this manner, external terminal **100** can start to use the one or more display regions **10a** for which the use request has been issued. After display region manager **32** permits the using of one or more display regions **10a**, display region manager **32** outputs, to display controller **33**, using information on the one or more display regions **10a** of which the using is permitted. Here, the using information on the one or more display regions **10a** includes, for each of the one or more display regions **10a**, display modes and presence or absence of an input and an output for information.

[Display Controller **33** of Display Control Device **30**]

[0062] In accordance with the using information on the one or more display regions **10a** obtained from display region manager **32**, display controller **33** issues an instruction to control event generator **35** and display screen generator **34**.

[0063] Specifically, when obtaining the using information on one or more display regions **10a**, display controller **33** issues the instruction to display screen generator **34** such that the one or more display regions **10a** indicated by the using information and the display mode according to the using information are executed. Here, the one or more display regions **10a** indicated by the using information are one or more display regions **10a** which display region manager **32** has permitted external terminal **100** to use. When display controller **33** executes the event driven mode being a display mode, display controller **33** issues an instruction to control event generator **35** and display screen generator **34**.

[Control Event Generator **35** of Display Control Device **30**]

[0064] For example, an application may be using display region **10a** to which the event driven mode is set in the display region table. Therefore, in a case where external terminal **100** is to use such display region **10a**, external terminal **100** does not know a timing to issue a display request and a use request. Thus, to inform external terminal **100** of a time period during which display region **10a** is allowed to be used, control event generator **35** issues an output request for outputting of display information to external terminal **100** when a preset event occurs. Here, the preset event is an event indicating that the application finishes the using of display region **10a**.

[0065] This will be described with a specific example. In a case where an application having a preferential usage right is not using display region **10a**, control event generator **35** obtains an instruction about the execution of the event driven mode from display controller **33**. That is, when the application having the preferential usage right finishes the using of display region **10a**, control event generator **35** issues an output request for outputting of display information to external terminal **100** in response to the instruction from display controller **33**. By obtaining the output request for outputting of display information, external terminal **100** can recognize a timing to use display region **10a** and thus can output the display information including a terminal-application image.

[Display Screen Generator **34** of Display Control Device **30**]

[0066] When display screen generator **34** obtains the instruction from display controller **33**, display screen generator **34** causes the plurality of displays **10** to display images.

[0067] Specifically, in a case where display region manager **32** permits external terminal **100** to use



one or more display regions **10a** selected by external terminal **100**, display screen generator **34** obtains, from external terminal **100**, display information including information to be displayed on one or more display regions **10a** selected by external terminal **100**. Here, as described above, the display information includes a terminal rendering command, a terminal-application image, and terminal text information including a text message from external terminal **100**. The terminal-application image and the terminal text information are an example of the information to be displayed on the one or more display regions **10a**. The terminal rendering command is a command to request display control system **2** to render terminal-application images to be displayed on the one or more display regions **10a** for which a use request has been issued to display control system **2**. [0068] Display screen generator **34** also obtains, from first storage **21**, a vehicle rendering command and a vehicle-application image of a vehicle application that corresponds to display region **10a** selected by external terminal **100**.

[0069] Thus, display screen generator **34** combines together the terminal-application image of the display information and the vehicle-application image of the vehicle application obtained from external terminal **100**. At this time, display screen generator **34** combines the terminal-application image and the vehicle-application image together in accordance with display modes that are preset to display region **10a**.

[0070] Here, as described above, the display modes each include the exclusive display mode, the message display mode, the overlap display mode, and the event driven mode.

[0071] The exclusive display mode is a mode in which external terminal **100** permitted to use display region **10a** can exclusively use display region **10a**. In the exclusive display mode, while external terminal **100** is using display region **10a**, another external terminal **100** is not permitted to interrupt.

[0072] The message display mode is a mode in which display region **10a** for displaying a character string can be used. In the message display mode, an image of a character string generated from the terminal text information is displayed on display region **10a**.

[0073] The overlap display mode is a mode in which display region **10a** capable of displaying an overlapped image can be used with a setting of a parameter relating to overlapping such as a transparency of the image. In the overlap display mode, a semitransparent image is displayed on display region **10a**.

[0074] The event driven mode is a mode in which display region **10a** is not allowed to be used while an application having a preferential usage right is using display region **10a**, and display region **10a** is allowed to be used when the application having a preferential usage right does not use display region **10a**.

[0075] In the event driven mode, when an application having a preferential usage right does not use display region **10a**, external terminal **100** can use display region **10a** in which the event driven mode is executed, by issuing an output request for outputting of display information to external terminal **100**. Specifically, in display control device **30**, a display mode associated with a preset event, that is, the event driven mode may be executed. In this case, display screen generator **34** obtains, external terminal **100**, the display information in response to the output request for outputting of the display information from control event generator **35**. Thus, display screen generator **34** renders an image of the obtained display information and the like in accordance with the event driven mode being a display mode. Display screen generator **34** displays rendered image information on the plurality of displays **10**.

[0076] In the event driven mode, even while external terminal **100** is using display region **10a**, an application having a preferential usage right can use display region **10a** being used by external terminal **100** in an interrupting manner. In this case, external terminal **100** is made to temporarily stop the using of display region **10a**.

[0077] In this manner, by combining the terminal-application image and the vehicle-application image together in accordance with one of these display modes, display screen generator **34** can

generate a single image in which the terminal-application image is rendered at a position corresponding to display region **10a** selected by external terminal **100**. By outputting the generated image to display **10** corresponding to display region **10a** which external terminal **100** is permitted to use, display screen generator **34** can cause display **10** to display the generated image.

<Processing Operation>

[0078] Next, processing operation of display control device **30**, display control system **2**, a display control method, and a non-transitory computer-readable recording medium having a computer program recorded thereon, in the present embodiment will be described.

Operation Example 1

[0079] In the present operation example, a processing operation from external terminal **100** issuing a display request and a use request to display control system **2** until external terminal **100** obtaining notification of permission for using of one or more display regions **10a** for which the use request has been issued will be described with reference to FIG. **4**.

[0080] FIG. **4** is a sequence diagram illustrating an example of a processing operation between display control system **2** and external terminal **100** according to the embodiment.

[0081] First, when external terminal **100** is to use one or more display regions **10a** included in a plurality of display regions **10a**, external terminal **100** issues a display request to display control system **2** (S11). Thus, the display request is issued to display control system **2** from external terminal **100**. When external terminal **100** issues the display request to display control system **2**, external terminal **100** can also notify display control system **2** of one or more identifiers of the one or more display regions **10a** selected to be used.

[0082] Next, display control system **2** obtains the display request from external terminal **100**. Specifically, display region manager **32** of display control system **2** obtains the display request from external terminal **100** via communicator **22**. When obtaining the display request, display region manager **32** reads a management table in second storage **31**. Display region manager **32** creates a display region table using the management table read (S21).

[0083] Specifically, display region manager **32** may extract only display region information items allowed to be used from a plurality of display region information items in the management table and may create a display region table that includes only the display region information items extracted. In this case, the display region table excludes display region information on one or more display regions **10a** that are not allowed to be used when a display request is issued. When display region manager **32** obtains, from external terminal **100**, one or more identifiers of one or more display regions **10a** selected to be used by external terminal **100** together with a display request, display region manager **32** may extract only one or more display regions **10a** selected to be used by external terminal **100** from a plurality of display regions **10a** in the management table and may create a display region table collectively indicating the display region information on the one or more display regions **10a** extracted.

[0084] Next, display region manager **32** outputs the display region table created to external terminal **100** via communicator **22** (S22).

[0085] Next, when external terminal **100** obtains the display region table from display control system **2**, external terminal **100** selects one or more display regions **10a** to be used from among the plurality of display regions **10a** based on the display region table (S12).

[0086] Next, external terminal **100** issues, to display control system **2**, a use request for permission to use the one or more display regions **10a** that are selected to use (S13).

[0087] Next, display control system **2** obtains the use request for permission to use the one or more display regions **10a** from external terminal **100**. Display region manager **32** of display control system **2** determines whether or not the one or more display regions **10a** requested by the use request are allowed to be used (S23). When determining that the one or more display regions **10a** are allowed to be used, display region manager **32** permits external terminal **100** for which the one or more display regions **10a** are determined to be allowed to be used, to use the one or more display

regions **10a** selected by external terminal **100**. The sequence diagram in FIG. 4 illustrates a case of permitting the using of the one or more display regions **10a** selected by external terminal **100** is exemplified.

[0088] Here, a case where display region manager **32** permits external terminal **100** to use one or more display regions **10a** will be specifically described.

[0089] First, display region manager **32** may determine that the one or more display regions **10a** are allowed to be used in response to a use request that is obtained first and may permit external terminal **100** that has issued the use request for which the one or more display regions **10a** are determined to be allowed to be used, to use the one or more display regions **10a** selected by external terminal **100**. That is, when display region manager **32** obtains the first use request after outputting the display region table created to external terminal **100**, display region manager **32** may permit external terminal **100** that has issued the first use request to use the one or more display regions **10a** selected by external terminal **100**. In this case, display region manager **32** may notify, via communicator **22**, external terminal **100** that has issued the first use request, of the permission for using.

[0090] Next, display region manager **32** may determine, in response to a use request, that the one or more display regions **10a** are allowed to be used only by external terminal **100** to which a usage right has been granted beforehand, and may permit external terminal **100** that has issued the use request for which the one or more display regions **10a** are determined to be allowed to be used, to use the one or more display regions **10a** selected by external terminal **100**. That is, when a use request has been issued by external terminal **100** to which the usage right has been granted, display region manager **32** may permit only external terminal **100** to which the usage right has been granted to use display region **10a** selected by external terminal **100**. In this case, display region manager **32** may notify, via communicator **22**, external terminal **100** to which the usage right has been granted, of the permission for using.

[0091] Next, display region manager **32** may determine that the one or more display regions **10a** are allowed to be used in response to a use request obtained from a device to which a preferential usage right has been granted beforehand, and may preferentially permit external terminal **100** that has issued the use request for which the one or more display regions **10a** are determined to be allowed to be used, to use the one or more display regions **10a** selected by the device to which the preferential usage right has been granted. That is, when a use request has been issued by the device having the preferential usage right, display region manager **32** may permit the device having the preferential usage right preferentially to use the one or more display regions **10a** selected by the device even when the one or more display regions **10a** have already been used by another device. In this case, display region manager **32** may notify, via communicator **22**, the device having the preferential usage right, of the permission for using. In addition, display region manager **32** temporarily does not allow display region **10a** used by the other device to be used. Thus, the other device is forced to temporarily stop the one or more display regions **10a** being used.

[0092] Next, the plurality of display regions **10a** may be associated with uses of the plurality of display regions **10a** beforehand. In this case, when display region manager **32** obtains a plurality of use requests, display region manager **32** may determine, in response to a use request for display region **10a** associated with a use that is most similar to a use of a display mode associated with display region **10a** among the plurality of use requests, that display region **10a** is allowed to be used, and may permit external terminal **100** that has issued the use request for which display region **10a** is determined to be allowed to be used, to use display region **10a** selected by external terminal **100**. That is, display region manager **32** checks uses of a plurality of external terminals **100** that have issued the use requests against a use that is associated beforehand with display region **10a** selected by external terminals **100**, thus extracting the use request for display region **10a** associated with a use that is most similar to the use of the display mode associated with display region **10a**. Display region manager **32** then may permit external terminal **100** that has issued the use request

for display region **10a** associated with a use that is most similar to the use of display region **10a** with which the display mode is associated, to use display region **10a** with which the display mode is associated. In this case, display region manager **32** may notify, via communicator **22**, external terminal **100** that has issued the use request for display region **10a** associated with a use that is most similar to the use of display region **10a** with which the display mode is associated, of the permission for using.

[0093] The use of each display region **10a** may be associated in accordance with the display mode. For example, as the use of display region **10a**, a use suitable for displaying a text message such as mail, a use suitable for displaying a warning sign, a use suitable for overlapped displaying, a use suitable for displaying an image of a navigation application, a use suitable for displaying an image of an audio application, and the like may be set to display region **10a** in accordance with the display mode. Display region **10a** associated with the exclusive display mode may be set such that display region **10a** can support all of the uses.

[0094] Next, the plurality of external terminals **100** each may have been preset with a priority degree based on which one or more display regions **10a** are preferentially determined to be allowed to be used. In this case, display region manager **32** may determine that one or more display regions **10a** are allowed to be used in response to a use request obtained from external terminal **100** having a highest priority degree among use requests obtained from the plurality of external terminals **100** and may permit external terminal **100** that has issued the use request for which the one or more display regions **10a** are determined to be allowed to be used, to use the one or more display regions **10a** selected by external terminal **100**. That is, display region manager **32** may permit external terminal **100** having the highest priority degree among priority degrees of the plurality of external terminals **100** that have issued the use requests, to use the one or more display regions **10a**. In this case, display region manager **32** may notify, via communicator **22**, external terminal **100** having the highest priority degree among the plurality of external terminals **100** that have issued the use requests, of the permission for using.

[0095] After permitting in this manner the using of the one or more display regions **10a** selected by external terminal **100**, display region manager **32** outputs using information on the one or more display regions **10a** of which the using is permitted, to display controller **33**.

[0096] On the other hand, when determining that the one or more display regions **10a** are not allowed to be used, display region manager **32** does not permit external terminal **100** for which the one or more display regions **10a** are determined not to be allowed to be used, to use the one or more display regions **10a** selected by external terminal **100**. In this case, display region manager **32** may finish the processing operation illustrated in the sequence diagram in FIG. 4. In addition, display region manager **32** may notify, via communicator **22**, external terminal **100** that the one or more display regions **10a** selected by external terminal **100** are not allowed to be used.

[0097] Next, display controller **33** obtains, from display region manager **32**, the using information on the one or more display regions **10a** which display region manager **32** permits external terminal **100** to use, thus executing preparation to start using (S24).

[0098] Specifically, when obtaining the using information on the one or more display regions **10a**, display controller **33** executes the preparation to start using by issuing an instruction to display screen generator **34** such that the one or more display regions **10a** indicated by the using information and the display mode according to the using information are executed. In addition, when display controller **33** executes the event driven mode being a display mode, display controller **33** executes the preparation to start using by issuing an instruction to control event generator **35** and display screen generator **34**.

[0099] Next, display region manager **32** notifies, external terminal **100** of the permission for using the one or more display regions **10a** selected by external terminal **100** (S25).

[0100] Thus, when external terminal **100** obtains the notification of the permission for using from display control system 2, external terminal **100** can start the using of the one or more display

regions **10a** that are selected by external terminal **100** and which external terminal **100** is permitted to use.

[0101] Next, the following operation example will describe a case where display information obtained is rendered in accordance with a display mode by display control device **30** executing the display mode after obtaining, display control system **2**, the notification of the permission for using the one or more display regions **10a** selected by external terminal **100**.

#### Operation Example 2: Exclusive Display Mode

[0102] In the present operation example, a processing operation in which display control system **2** executes the exclusive display mode as a display mode will be described with reference to FIG. **5**.

[0103] FIG. **5** is a sequence diagram illustrating an example of a processing operation between display control system **2** executing the exclusive display mode and external terminal **100**.

[0104] In the present operation example, a case where a terminal-application image is displayed on display region A will be exemplified. Display controller **33** executes the exclusive display mode, which is a display mode that has been preset to display region A, that is, a display mode that has been associated with display region A.

[0105] First, external terminal **100** obtains a notification of permission for using from display control system **2** (S31).

[0106] Next, external terminal **100** transmits, to display control system **2**, display information that includes a terminal-application image to be displayed on display region A selected by external terminal **100** in S12 of Operation example 1 and includes a terminal rendering command (S32).

[0107] Next, display screen generator **34** obtains the display information from external terminal **100** via communicator **22**. Display screen generator **34** also obtains, from first storage **21**, a vehicle rendering command and a vehicle-application image of a vehicle application that corresponds to display region A selected by external terminal **100**. Thus, display screen generator **34** generates an image by combining together the terminal-application image of the display information and the vehicle-application image from the vehicle application obtained from external terminal **100** (S33). That is, by combining the terminal-application image and the vehicle-application image together, display screen generator **34** can generate a single image in which the terminal-application image is rendered at a position corresponding to display region A selected by external terminal **100**.

[0108] Next, display screen generator **34** outputs the generated single image to display **10** corresponding to display region A which external terminal **100** is permitted to use (S34). Thus, the generated single image is displayed as the display data on display **10**, and the terminal-application image rendered is displayed as the display data on display region A which external terminal **100** is permitted to use.

[0109] Next, display screen generator **34** notifies external terminal **100** that the display of the images has been completed by rendering (S35).

[0110] In a case where external terminal **100** is to further display another terminal-application image, external terminal **100** may repeat the processing of step S32 and subsequent steps (S36).

#### Operation Example 3: Message Display Mode

[0111] In the present operation example, a processing operation in which display control system **2** executes the message display mode as a display mode will be described with reference to FIG. **6**.

[0112] FIG. **6** is a sequence diagram illustrating an example of a processing operation between display control system **2** executing the message display mode and external terminal **100**.

[0113] In the present operation example, a case where terminal text information is displayed on display region C will be exemplified. Display controller **33** executes the message display mode, which is a display mode that has been preset to display region C, that is, a display mode that has been associated with display region C.

[0114] First, external terminal **100** obtains a notification of permission for using from display control system **2** (S41).

[0115] Next, external terminal **100** transmits, to display control system **2**, display information that

includes the terminal text information to be displayed on display region C selected by external terminal **100** in S12 of Operation example 1 and includes a terminal rendering command (S42). Here, the terminal text information includes a text message and a limit on the number of characters. [0116] Next, display screen generator **34** receives the display information from external terminal **100** via communicator **22**. Display screen generator **34** also obtains, from first storage **21**, a vehicle rendering command and a vehicle-application image of a vehicle application that corresponds to display region C selected by external terminal **100**. Thus, display screen generator **34** generates an image by combining together the terminal text information of the display information and the vehicle-application image from the vehicle application obtained from external terminal **100** (S43). That is, by combining the terminal text information and the vehicle-application image together, display screen generator **34** can generate a single image in which the terminal text information is rendered at a position corresponding to display region C selected by external terminal **100**. [0117] Next, display screen generator **34** outputs the generated single image to display **10** corresponding to display region C which external terminal **100** is permitted to use (S44). Thus, the generated single image is displayed as the display data on display **10**, and the terminal text information rendered is displayed as the display data on display region C which external terminal **100** is permitted to use.

[0118] Next, display screen generator **34** notifies external terminal **100** that the display of the images has been completed by rendering (S45).

[0119] In a case where external terminal **100** is to further display another terminal text information item, external terminal **100** may repeat the processing of step S42 and subsequent steps (S46).

#### Operation Example 4: Overlap Display Mode

[0120] In the present operation example, a processing operation in which display control system **2** executes the overlap display mode as a display mode will be described with reference to FIG. 7.

[0121] FIG. 7 is a sequence diagram illustrating an example of a processing operation between display control system **2** executing the overlap display mode and external terminal **100**.

[0122] In the present operation example, a case where a terminal-application image is displayed on display region E will be exemplified. Display controller **33** executes the overlap display mode, which is a display mode that has been preset to display region E, that is, a display mode that has been associated with display region E.

[0123] First, external terminal **100** obtains a notification of permission for using from display control system **2** (S51).

[0124] Next, external terminal **100** outputs, to display control system **2**, a parameter relating to overlapping for setting a transparency and the like of the terminal-application image to be displayed on display region E (S52).

[0125] Next, display screen generator **34** obtains the parameter relating to overlapping via communicator **22**. At this time, display screen generator **34** determines whether or not the parameter relating to overlapping is executable. When the parameter is executable, display screen generator **34** sets the parameter relating to overlapping obtained (S53). When the parameter is not executable, display screen generator **34** may notify, via communicator **22**, external terminal **100** that the parameter is not executable.

[0126] Next, when the parameter relating to overlapping is successfully set, display screen generator **34** notifies external terminal **100** that the setting of the parameter is completed (S54).

[0127] Next, external terminal **100** transmits, to display control system **2**, display information that includes a terminal-application image to be displayed on display region E selected by external terminal **100** in S12 of Operation example 1 and includes a terminal rendering command (S55).

[0128] Next, display screen generator **34** obtains the display information from external terminal **100** via communicator **22**. Display screen generator **34** also obtains, from first storage **21**, a vehicle rendering command and a vehicle-application image of a vehicle application that corresponds to display region E selected by external terminal **100**. In addition, display screen generator **34** makes

the terminal-application image of the display information obtained from external terminal **100** semitransparent based on the parameter relating to overlapping. Thus, display screen generator **34** generates an image by combining together the terminal-application image made semitransparent based on the parameter and the vehicle-application image from the vehicle application (S56). That is, by combining the semitransparent terminal-application image and the vehicle-application image together, display screen generator **34** can generate a single image in which the terminal-application image made semitransparent is rendered at a position corresponding to display region E selected by external terminal **100**.

[0129] Next, display screen generator **34** outputs the generated single image to display **10** corresponding to display region E which external terminal **100** is permitted to use (S57). Thus, the generated single image is displayed as the display data on display **10**, and the semitransparent terminal-application image rendered is displayed as the display data on display region E which external terminal **100** is permitted to use.

[0130] Next, display screen generator **34** notifies external terminal **100** that the display of the images has been completed by rendering (S58).

[0131] In a case where external terminal **100** is to further display another terminal-application image, external terminal **100** may repeat the processing of step S55 and subsequent steps (S59).

[0132] In the overlap display mode, display control system **2** may set the parameter relating to overlapping. In this case, processing in step S52 may be omitted.

#### Operation Example 5: Event Driven Mode

[0133] In the present operation example, a processing operation in which display control system **2** executes the event driven mode as a display mode will be described with reference to FIG. **8**.

[0134] FIG. **8** is a sequence diagram illustrating an example of a processing operation between display control system **2** executing the event driven mode and external terminal **100**.

[0135] In the present operation example, a case where a terminal-application image is displayed on display regions B and D will be exemplified. Display controller **33** executes the event driven mode, which is a display mode that has been preset to display regions B and D, that is, a display mode that has been associated with display regions B and D.

[0136] First, external terminal **100** obtains a notification of permission for using from display control system **2** (S61).

[0137] Next, when a preset event occurs, display controller **33** of display control system **2** issues an instruction to control event generator **35**. The preset event is an event that occurs when external terminal **100** becomes able to use display regions B and D. Display controller **33** obtains such an event from a vehicle application or the like. When obtaining the instruction from display controller **33**, control event generator **35** issues an output request for outputting of display information to external terminal **100** (S62).

[0138] Next, when obtaining the notification of permission for using and the output request for outputting of display information, external terminal **100** transmits, to display control system **2**, display information that includes a terminal-application image to be displayed on display regions B and D selected by external terminal **100** in S12 of Operation example 1 and includes a terminal rendering command (S63).

[0139] Next, display screen generator **34** obtains the display information from external terminal **100** via communicator **22**. Display screen generator **34** also obtains, from first storage **21**, a vehicle rendering command and a vehicle-application image of a vehicle application that corresponds to display regions B and D selected by external terminal **100**. Thus, display screen generator **34** generates an image by combining together the terminal-application image of the display information and the vehicle-application image from the vehicle application obtained from external terminal **100** (S64). That is, by combining the terminal-application image and the vehicle-application image together, display screen generator **34** can generate a single image in which the terminal-application image is rendered at positions corresponding to display regions B and D

selected by external terminal **100**.

[0140] Next, display screen generator **34** outputs the generated single image to display **10** corresponding to display regions B and D which external terminal **100** is permitted to use (S65). Thus, the generated single image is displayed as the display data on display **10**, and the terminal-application image rendered is displayed as the display data on display regions B and D which external terminal **100** is permitted to use.

[0141] Next, display screen generator **34** notifies external terminal **100** that the display of the images has been completed by rendering (S66).

[0142] In a case where external terminal **100** is to further display another terminal-application image, external terminal **100** may repeat the processing of step S63 and subsequent steps (S67).

[0143] For display regions B and D being used by external terminal **100**, there may be an event occurring for temporarily using by a vehicle application. The vehicle application is an application to which a preferential usage right has been granted and thus can preferentially use display regions B and D being already used. Thus, the vehicle application displays an image corresponding to the event occurring on display regions B and D, and external terminal **100** stops displaying the terminal-application image on display regions B and D. Here, the image corresponding to the event is, for example, a warning sign in vehicle **1**.

[0144] Next, display control system **2** notifies external terminal **100** of a rendering pending request (S68).

[0145] Next, when external terminal **100** obtains the rendering pending request, external terminal **100** enters a waiting state for using display regions B and D. When the waiting state for using is canceled, external terminal **100** can display the terminal-application image on display regions B and D again.

#### Actions and Effects

[0146] Next, actions and effects of display control device **30**, display control system **2**, a display control method, and a non-transitory computer-readable recording medium having a computer program stored thereon, in the present embodiment will be described.

[0147] In the on-vehicle system according to patent document 1, displaying application software information on a display area depends on an application of the mobile terminal device. Therefore, the on-vehicle system supports limited applications, which may cause the displaying to be performed improperly. In addition, the on-vehicle system allocates application software information items to specific display areas. Therefore, rendering can be performed only on the display areas that are one-sidedly determined by the on-vehicle system, limiting a degree of freedom in expression in the plurality of display areas in a car interior.

[0148] In addition, in the apparatus according to Patent Literature 2, settings for displaying information on regions in one or more displays are made by a user. This requires the user to make detailed settings and the like of the region to display the information, thus decreasing operability.

[0149] As described above, display control device **30** according to Embodiment for use in vehicle **1** provided with a plurality of displays **10** each including one or more display regions **10a** includes: display region manager **32** that manages a plurality of display regions **10a** that are the one or more display regions **10a** included in each of the plurality of displays **10**, wherein when display region manager **32** obtains, from external terminal **100**, a display request for displaying display data on display region **10a** included in the plurality of display regions **10a**, display region manager **32**: creates a display region table collectively indicating display region information on at least one display region **10a** among the plurality of display regions **10a**; and [0150] outputs the display region table created to external terminal **100**.

[0151] Accordingly, display region manager **32** can create the display region table collectively indicating information on at least one display region **10a** and output the display region table to external terminal **100**. Therefore, external terminal **100** can select desired one or more display regions **10a** from display region information indicated by the display region table. Thus, the one or



more display regions **10a** selected by external terminal **100** can display the display data from external terminal **100** (the terminal-application image, the terminal text information, etc.).

[0152] According to Embodiment, the user does not need to perform setting for displaying data on a desired display region by operating the display control device and the external terminal, which is necessary in the related arts. Therefore, display control device **30** according to Embodiment prevents the user's operability from being deteriorated.

[0153] Therefore, display control device **30** can display the display data from external terminal **100** on one or more display regions **10a** of vehicle **1** appropriately and freely and can prevent a decrease in operability for a user.

[0154] Furthermore, display control system **2** according to Embodiment includes: display control device **30** and plurality of displays **10**.

[0155] Display control system **2** having this configuration also exhibits the same actions and effects as described above.

[0156] A display control method according to Embodiment for use in vehicle **1** provided with a plurality of displays **10** each including one or more display regions **10a** includes: managing a plurality of display regions **10a** that are the one or more display regions **10a** included in each of plurality of displays **10**; when a display request for displaying of display data on display region **10a** included in the plurality of display regions **10a** is obtained from external terminal **100**, creating a display region table collectively indicating display region information on at least one display region **10a** among the plurality of display regions **10a**; and outputting the display region table created to external terminal **100**.

[0157] This display control method also exhibits the same actions and effects as described above.

[0158] The recording medium according to Embodiment is a non-transitory computer-readable recording medium having a computer program recorded thereon for causing the computer to execute the display control method.

[0159] This recording medium also exhibits the same actions and effects as described above.

[0160] In display control device **30** according to Embodiment, display region manager **32**: extracts only display region **10a** allowed to be used from the plurality of display regions **10a**; and creates the display region table collectively indicating only the display region information on display region **10a** extracted.

[0161] Accordingly, display region **10a** allowed to be used can be extracted, and thus it is possible to create a display region table in which one or more display regions **10a** not allowed to be used are not included. That is, external terminal **100** can use display region information on only display region **10a** allowed to be used, and thus it is possible for display control device **30** to prevent occurrence of such a failure that external terminal **100** selects one or more display regions **10a** not allowed to be used. As a result, display control device **30** can prevent occurrence of unnecessary communication between display control device **30** and external terminal **100**.

[0162] In display control device **30** according to Embodiment, external terminal **100** selects display region **10a** to be used from among the plurality of display regions **10a** based on the display region table, and issues a use request for permission to use display region **10a** selected, and display region manager **32**: obtains, from external terminal **100**, the use request for display region **10a** selected by external terminal **100**; determines whether or not display region **10a** requested by the use request is allowed to be used; when determining that display region **10a** is allowed to be used, permits external terminal **100** for which display region **10a** is determined to be allowed to be used, to use display region **10a** selected by external terminal **100**; and when determining that display region **10a** is not allowed to be used, prohibits external terminal **100** for which display region **10a** is not determined to be allowed to be used, from using display region **10a** selected by external terminal **100**.

[0163] Accordingly, display region manager **32** can determine whether or not display region **10a** selected by external terminal **100** is allowed to be used, in response to a use request for permission

to use display region **10a** selected by external terminal **100**. For example, even when a plurality of external terminals **100** issue use requests for permission to use the same display region **10a**, whether or not the same display region **10a** is allowed to be used can be determined in an above-described manner. Therefore, display control device **30** can prevent display region manager **32** from permitting the plurality of external terminals **100** to use the same display region **10a**.

[0164] In display control device **30** according to Embodiment, the display region information includes, for each of the at least one display region **10a**, an identifier, a position, a size, a display mode, presence or absence of an input and an output for information, and a use condition.

[0165] Accordingly, external terminal **100** can select display data that is suitable for display region information in the display region table. Therefore, display control device **30** can cause at least one display region **10a** to display the display data suitable for the display region information.

[0166] In display control device **30** according to Embodiment, display region manager **32**: creates the display region table collectively indicating the display region information on display region **10a** selected to be used by external terminal **100** from among the plurality of display regions **10a**; and outputs the display region table created to external terminal **100**.

[0167] Accordingly, display region manager **32** can create a display region table including display region **10a** selected to be used by external terminal **100**. Therefore, external terminal **100** can select display region **10a** to be used from the display region table. As a result, display control device **30** can display the display data on display region **10a** selected to be used by external terminal **100**.

[0168] In display control device **30** according to Embodiment, display region manager **32**: determines that display region **10a** is allowed to be used in response to the use request that is obtained first; and permits external terminal **100** that has issued the use request for which display region **10a** is determined to be allowed to be used, to use display region **10a** selected by external terminal **100**.

[0169] Accordingly, it is possible to permit only external terminal **100** that has issued the use request obtained first to use display region **10a**. Therefore, even when a plurality of external terminals **100** issue use requests for permission to use the same display region **10a**, display control device **30** can prevent display region manager **32** from permitting the plurality of external terminals **100** to use the same display region **10a**.

[0170] In display control device **30** according to Embodiment, display region manager **32**: determines, in response to the use request, that display region **10a** is allowed to be used only by external terminal **100** to which a usage right has been granted beforehand; and permits external terminal **100** that has issued the use request for which display region **10a** is determined to be allowed to be used, to use display region **10a** selected by external terminal **100**.

[0171] Accordingly, in response to the use request, only external terminal **100** to which the usage right has been granted beforehand is permitted to use display region **10a**, and thus it is possible to prevent external terminal **100** to which no usage right has been granted from using display region **10a**. Therefore, display control device **30** can prevent external terminal **100** from using display region **10a** when, for example, a warning sign, a direction indicator, or the like of vehicle **1** is displayed. Therefore, display control device **30** can keep safety of vehicle **1** being traveling.

[0172] In display control device **30** according to Embodiment, display region manager **32**: determines that display region **10a** is allowed to be used in response to the use request obtained from a device to which a preferential usage right has been granted beforehand; and preferentially permits the device that has issued the use request for which display region **10a** is determined to be allowed to be used, to use display region **10a** selected by the device.

[0173] Accordingly, even when external terminal **100** has already used display region **10a**, the device to which the preferential usage right has been granted can preferentially use display region **10a** that is being used by external terminal **100**. Therefore, when an event relating to vehicle **1** occurs, display control device **30** can preferentially display data corresponding to the event on display region **10a**. Therefore, display control device **30** can keep safety of vehicle **1** being

traveling.

[0174] In display control device **30** according to Embodiment, when display region manager **32** obtains a plurality of use requests each being the use request, display region manager **32**: determines, in response to a use request for display region **10a** associated with a use that is most similar to a use of a display mode associated with display region **10a** among the plurality of use requests, that display region **10a** is allowed to be used; and permits external terminal **100** that has issued the use request for which display region **10a** is determined to be allowed to be used, to use display region **10a** selected by external terminal **100**.

[0175] Accordingly, display region **10a** suitable for external terminal **100** can be used. Therefore, display control device **30** can cause at least one display region **10a** to display the display data suitable for the display region information.

[0176] In display control device **30** according to Embodiment, a plurality of external terminals **100** each being external terminal **100** each have been preset with a priority degree based on which display region **10a** is preferentially determined to be allowed to be used, and display region manager **32**: determines, in response to a use request obtained from external terminal **100** having a highest priority degree among a plurality of use requests obtained from the plurality of external terminals **100**, that display region **10a** is allowed to be used; and permits external terminal **100** that has issued the use request for which display region **10a** is determined to be allowed to be used, to use display region **10a** selected by external terminal **100**.

[0177] Accordingly, in response to the use request obtained from external terminal **100** having the highest priority degree among the use requests obtained from the plurality of external terminals **100**, it is possible to permit external terminal **100** having the highest priority degree to use display region **10a**. Therefore, even when a plurality of external terminals **100** issue use requests for permission to use the same display region **10a**, display control device **30** can prevent display region manager **32** from permitting the plurality of external terminals **100** to use the same display region **10a**.

[0178] Display control device **30** according to Embodiment further includes: display screen generator **34** that causes the plurality of displays **10** to display images, wherein when display region manager **32** permits the external device to use display region **10a** selected by the external device, display screen generator **34**: obtains, from external terminal **100**, display information including information to be displayed on display region **10a** selected by external terminal **100**; renders an image of the display information obtained in a display mode associated with display region **10a**, by executing the display mode; and causes display **10** corresponding to display region **10a** which the external device is permitted to use among the plurality of displays **10**, to display the image rendered.

[0179] Accordingly, an image suitable for display region **10a** can be generated by executing the display mode. Therefore, display control device **30** can cause at least one display region **10a** to display the display data suitable for the display region information.

[0180] Display control device **30** according to Embodiment further includes: control event generator **35** that issues an output request for outputting of the display information to external terminal **100**, when a preset event occurs, wherein display screen generator **34**: obtains, from external terminal **100**, the display information based on the output request; renders an image of the display information obtained in a display mode associated with the preset event, by executing the display mode; and causes display **10** corresponding to display region **10a** which the external device is permitted to use, to display the image rendered.

[0181] Accordingly, external terminal **100** is notified of the output request, and thus external terminal **100** can use display region **10a**. Therefore, display control device **30** can improve convenience when external terminal **100** uses display region **10a**.

Other Embodiments

[0182] Although the display control device, the display control system, the display control method,

and the recording medium according to one or more aspects of the present disclosure has been described based on an embodiment, the present disclosure is not limited to this embodiment. Those skilled in the art will readily appreciate that embodiments arrived at by making various modifications to the above embodiment without materially departing from the scope of the present disclosure may be included within one or more aspects of the present disclosure.

[0183] For example, in the display control device, the display control system, the display control method, and the recording medium according to the embodiment described above, the display mode includes the exclusive display mode, the message display mode, the overlap display mode, and the event driven mode. However, it may suffice that these modes are selectively included. That is, the display mode need not necessarily include these four modes. In this case, when the display mode does not include the event driven mode, the display control device and the display control system need not include the control event generator. That is, the control event generator is not an essential constituent component of the display control device and the display control system.

[0184] Each of the elements in each of the display region manager, the display controller, control event generator, and display screen generator which are included in the display control device and the display control system according to Embodiment may be configured in the form of an exclusive hardware product, or may be realized by executing a software program suitable for the element. Each of the elements may be realized by means of a program executing unit, such as a Central Processing Unit (CPU) or a processor, reading and executing the software program recorded on a recording medium such as a hard disk or semiconductor memory.

[0185] Furthermore, all the numerical values indicated in the above are examples for explaining the present disclosure in detail, and do not limit the present disclosure.

[0186] The dividing of the functional blocks in each of the block diagrams is one example. It is possible that a plurality of functional blocks are implemented into a single functional block, that a single functional block is divided into a plurality of functional blocks, and that a function executed by a functional block is partially executed by another functional block. Furthermore, similar functions of a plurality of functional blocks may be executed by a single hardware or software in parallel or by time division.

[0187] An order of performing the steps in each of the flowcharts is an example for explaining the present disclosure in detail. The steps may be performed in different orders. Different steps among the steps may be performed simultaneously, in other words, in parallel.

[0188] Those skilled in the art will readily appreciate that embodiments arrived at by making various modifications to the above embodiments or embodiments arrived at by selectively combining elements disclosed in the above embodiments without materially departing from the scope of the present disclosure may be included within one or more aspects of the present disclosure.

[0189] While various embodiments have been described herein above, it is to be appreciated that various changes in form and detail may be made without departing from the spirit and scope of the present disclosure as presently or hereafter claimed.

#### INDUSTRIAL APPLICABILITY

[0190] The present disclosure is applicable to, for example, a vehicle, a device other than a vehicle, and a system that are provided with a plurality of display devices, and to a recording medium.

## Claims

1. A display control device for use in a vehicle provided with a display that includes a plurality of display regions, the display control device comprising: a display region manager that manages the plurality of display regions, wherein when the display region manager obtains, from an external terminal, a display request for displaying display data on a display region included in the plurality of display regions, the display region manager: creates a display region table collectively indicating

display region information on at least one display region among the plurality of display regions; and outputs the display region table to the external terminal.

2. The display control device according to claim 1, wherein the display region manager: extracts only a display region allowed to be used from the plurality of display regions; and creates the display region table collectively indicating only the display region information on the display region extracted.

3. The display control device according to claim 1, wherein the external terminal selects a display region to be used from among the plurality of display regions based on the display region table, and issues a use request for permission to use the display region selected, and the display region manager: obtains, from the external terminal, the use request for the display region selected by the external terminal; determines whether or not the display region requested by the use request is allowed to be used; when determining that the display region is allowed to be used, permits the external terminal for which the display region is determined to be allowed to be used, to use the display region selected by the external terminal; and when determining that the display region is not allowed to be used, prohibits the external terminal for which the display region is not determined to be allowed to be used, from using the display region selected by the external terminal.

4. The display control device according to claim 1, wherein the display region information includes, for each of the at least one display region, an identifier, a position, a size, a display mode, presence or absence of an input and an output for information, and a use condition.

5. The display control device according to claim 1, wherein the display region manager: creates the display region table collectively indicating the display region information on a display region selected to be used by the external terminal from among the plurality of display regions; and outputs the display region table created to the external terminal.

6. The display control device according to claim 3, wherein the display region manager: determines that the display region is allowed to be used in response to the use request that is obtained first; and permits the external terminal that has issued the use request for which the display region is determined to be allowed to be used, to use the display region selected by the external terminal.

7. The display control device according to claim 3, wherein the display region manager: determines, in response to the use request, that the display region is allowed to be used only by the external terminal to which a usage right has been granted beforehand; and permits the external terminal that has issued the use request for which the display region is determined to be allowed to be used, to use the display region selected by the external terminal.

8. The display control device according to claim 3, wherein the display region manager: determines that the display region is allowed to be used in response to the use request obtained from a device to which a preferential usage right has been granted beforehand; and preferentially permits the device that has issued the use request for which the display region is determined to be allowed to be used, to use the display region selected by the device.

9. The display control device according to claim 3, wherein when the display region manager obtains a plurality of use requests each being the use request, the display region manager: determines, in response to a use request for a display region associated with a use that is most similar to a use of a display mode associated with the display region among the plurality of use requests, that the display region is allowed to be used; and permits the external terminal that has issued the use request for which the display region is determined to be allowed to be used, to use the display region selected by the external terminal.

10. The display control device according to claim 3, wherein a plurality of external terminals each being the external terminal each have been preset with a priority degree based on which the display region is preferentially determined to be allowed to be used, and the display region manager: determines, in response to a use request obtained from an external terminal having a highest priority degree among a plurality of use requests obtained from the plurality of external terminals,

that the display region is allowed to be used; and permits the external terminal that has issued the use request for which the display region is determined to be allowed to be used, to use the display region selected by the external terminal.

**11.** The display control device according to claim 1, further comprising: a display screen generator that causes the plurality of displays to display images, wherein when the display region manager permits the external device to use the display region selected by the external device, the display screen generator: obtains, from the external terminal, display information including information to be displayed on the display region selected by the external terminal; renders an image of the display information obtained in a display mode associated with the display region, by executing the display mode; and causes a display corresponding to the display region which the external device is permitted to use among the plurality of displays, to display the image rendered.

**12.** The display control device according to claim 11, further comprising: a control event generator that issues an output request for outputting of the display information to the external terminal, when a preset event occurs, wherein the display screen generator: obtains, from the external terminal, the display information based on the output request; renders an image of the display information obtained in a display mode associated with the preset event, by executing the display mode; and causes the display corresponding to the display region which the external device is permitted to use, to display the image rendered.

**13.** A display control system comprising: the display control device according to claim 1; and the display.

**14.** A display control method for use in a vehicle provided with a display that includes a plurality of display regions, the display control method comprising: managing the plurality of display regions; when a display request for displaying of display data on a display region included in the plurality of display regions, creating a display region table collectively indicating display region information on at least one display region among the plurality of display regions; and outputting the display region table to the external terminal.

**15.** A non-transitory computer-readable recording medium for use in a computer, the recording medium having a computer program recorded thereon for causing the computer to execute the display control method according to claim 14.

**16.** The display control device according to claim 1, wherein the display comprises a plurality of displays, and wherein each of the displays includes a respective plurality of display regions.

**17.** A display control device for use in a vehicle provided with a plurality of displays each including one or more display regions, the display control device comprising: a display region manager that manages a plurality of display regions that are the one or more display regions included in each of the plurality of displays, wherein the display region manager: creates a display region table collectively indicating display region information on at least one display region among the plurality of display regions; and outputs the display region table to the external terminal.

**18.** The display control device according to claim 17, wherein the display region manager: extracts only a display region allowed to be used from the plurality of display regions; and creates the display region table collectively indicating only the display region information on the display region extracted.

**19.** The display control device according to claim 17, wherein the external terminal selects a display region to be used from among the plurality of display regions based on the display region table, and issues a use request for permission to use the display region selected, and the display region manager: obtains, from the external terminal, the use request for the display region selected by the external terminal; determines whether or not the display region requested by the use request is allowed to be used; when determining that the display region is allowed to be used, permits the external terminal for which the display region is determined to be allowed to be used, to use the display region selected by the external terminal; and when determining that the display region is not allowed to be used, prohibits the external terminal for which the display region is not

determined to be allowed to be used, from using the display region selected by the external terminal.

**20.** The display control device according to claim 19, wherein the display region manager: determines that the display region is allowed to be used in response to the use request that is obtained first; and permits the external terminal that has issued the use request for which the display region is determined to be allowed to be used, to use the display region selected by the external terminal.

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